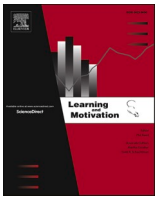
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Disclosing Chinese college students' flow experience in GenAI-assisted informal digital learning of english: A self-determination theory perspective

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ABSTRACT

The role of flow experience in enhancing GenAI-assisted informal digital learning of English (GAI-IDLE) has recently gained increasing attention. However, the mechanisms underlying the achievement of flow experience in this context remain underexplored. Grounded in Self-Determination Theory (SDT), this study investigates how GAI-IDLE influences flow experience through the lens of three basic psychological needs: autonomy, competence, and relatedness. Specifically, we aim to examine the mediating effects of these psychological needs on the relationship between GAI-IDLE and flow experience. To achieve this, we conducted a survey among 333 English as a Foreign Language (EFL) learners from various colleges in China. Using Partial Least Squares Structural Equation Modeling (PLS-SEM) with Smart PLS 4.0 software, our results show that GenAI literacy and English proficiency positively and directly impacted flow experience, while gender had no direct effect. GAI-IDLE directly predicted flow experience and positively influenced autonomy and relatedness, which in turn directly enhanced flow experience. Thus, GAI-IDLE indirectly affected flow experience through the mediation of autonomy and relatedness. However, contrary to SDT, GAI-IDLE did not directly affect competence, and competence did not directly influence flow experience, resulting in an insignificant mediating role for competence. We discuss potential explanations for these findings and conclude with educational implications.

1. Introduction

Learning English is crucial as it serves as a lingua franca, greatly facilitating international communication. In the digital age, with approximately 60 % of online content presented in English and its widespread use in educational settings (Kramer, 2023), English proficiency is the key to accessing a vast amount of information and study opportunities abroad. Moreover, in an increasingly globalized job market, English plays a vital role in academic collaboration, research dissemination, and career advancement (Habibie & Flowerdew, 2023). Fluency in English often gives individuals a competitive edge in employment (Peltokorpi, 2023). For students learning English as a Foreign Language (EFL), formal instruction, while systematic and standardized, has some drawbacks (Odo, 2020).

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The traditional classroom environment is often teacher-centered, which may limit student engagement and initiative. The fixed curriculum may not meet the diverse needs of individual learners, and the learning scenarios are mainly confined to the classroom, making it difficult for students to apply what they have learned in real-life situations.

Informal learning has emerged as an alternative approach to address these issues. Informal digital learning of English (IDLE) refers to the process of acquiring English language skills through digital technologies and online resources outside the traditional classroom (Lee, 2019). This learning approach is characterized by its flexibility, learner autonomy, and diverse range of content available (Liu et al., 2024). For instance, EFL learners can explore various online platforms that host numerous courses offered by esteemed domestic and international universities on MOOC platforms (Cha & So, 2020). These platforms empower learners to choose courses that align with their individual schedules and specific educational needs (Y. Wang et al., 2024). Additionally, language learning applications such as Duolingo provide engaging English courses that cover multiple facets of language learning, including vocabulary, grammar, listening, and speaking (Derakhshan et al., 2024; Ding & Wang, 2025; García Botero et al., 2019). These applications facilitate effective study practices by allowing users to take advantage of fragmented time.

However, traditional IDLE resources often present content in a rigid, fixed sequence, similar to conventional online courses (Guo & Lee, 2023). This inflexibility makes it difficult to tailor learning pathways and materials to individual learners' language proficiency, objectives, and preferences, resulting in a lack of personalized approaches. Additionally, learners typically engage with content in a one-dimensional manner, mainly through video lectures or reading materials, which limits interactive, real-time communication opportunities (Soyoof et al., 2023). These constraints can hinder the overall effectiveness of traditional IDLE methods. The rise of generative artificial intelligence (GenAI) offers a promising solution to the challenges faced by traditional IDLE (Guo & Wang, 2024; Rezai, Soyoof, & Reynolds, 2024; Wu, 2025). With its advanced data analysis and adaptive capabilities, GenAI can assess a wide range of factors related to learners' language skills, study habits, and knowledge acquisition (Pan & Wang, 2025; Wang & Xue, 2024; Wu & Pan, 2025). This enables the creation of customized learning paths and materials, delivering precise feedback and tailored recommendations (Derakhshan & Ghiasvand, 2024; Huang et al., 2024).

Additionally, GenAI can simulate real-world communication situations, allowing for seamless, real-time interactions with learners (Derakhshan et al., 2025; Wang et al., 2025; Wu, Wang, et al., 2024). It guides conversations based on user input, providing instant language assistance and corrections (Derakhshan, 2025; Rezai, Soyoof, & Lee Reynolds, 2024; Wang, 2017). For example, Doubao, a newly introduced conversational GenAI, allows EFL learners to engage in various scenarios, such as job interviews, IELTS speaking tests, and casual conversations. The integration of GenAI assisted-IDLE (GAI-IDLE) into education enhances interactivity and intelligence, leading to improved language proficiency. Recent studies on after-class interventions have confirmed these benefits. Zhang, Zou and Cheng (2025) developed a GPT-4-based logic learning bot for 40 EFL students, which led to significant improvements in their English argumentative writing skills. Similarly, C. Wang et al. (2024) explored the effects of conversational GenAI on the communicative abilities of 66 EFL learners, finding positive results. A recent meta-analysis by Guan et al. (2024) also emphasized that GenAI significantly improves English proficiency in informal settings.

Self-Determination Theory (SDT) (Deci & Ryan, 2015) is a well-established theoretical framework that focuses on basic psychological needs. It posits that human beings have three innate psychological needs: autonomy, competence, and relatedness. When these needs are satisfied, individuals are more likely to be motivated, engaged, and experience well-being. In the context of GAI-IDLE, SDT can provide a valuable lens to understand the effectiveness of this learning approach. By allowing students to choose their learning paths and pace, GenAI enhances their sense of autonomy, which is crucial for self-directed learning. Regarding competence, GenAI can accurately assess students' English proficiency, offer tailored tasks, and provide timely feedback, helping students achieve a sense of accomplishment. For relatedness, GenAI can create real-life interaction scenarios, meeting students' need for social connection within the learning process. Several researchers have emphasized the importance of SDT in educational research, and applying it to GAI-IDLE can offer a deeper understanding of how this learning method impacts learners' psychological states and learning outcomes (Gao et al., 2025; Jiang et al., 2025; Wang, Pan & Wang, 2024).

Several studies have investigated the psychological factors related to GAI-IDLE. For example, Ericsson and Johansson (2023) conducted a six-month observation of 22 secondary students using conversational GenAI for speaking practice. The results revealed that students felt comfortable, engaged, and immersed while interacting with GenAI, and they perceived improvements in their oral English over time. Similarly, in an experimental study, Zhang et al. (2024) found that among 133 EFL learners, speaking anxiety decreased, while enjoyment and willingness to communicate increased under GAI-IDLE. In another after-class experimental study, Liu (2024) examined 486 EFL learners and found that the experimental group showed significant improvements in both overall English proficiency and motivation for learning, following GenAI-enhanced gamification. These studies highlight several psychological factors that contribute to the effectiveness of GAI-IDLE, including enhanced enjoyment, increased motivation, and greater concentration. (Zhang et al., 2025) further explored these factors, conceptualizing them as "flow experience". Flow, a concept introduced by Csikszentmihalyi (1990), refers to an optimal state of complete immersion in an enjoyable and engaging activity.

In the context of GAI-IDLE, the combination of SDT and the concept of flow can provide a more comprehensive theoretical basis. SDT explains the fundamental psychological needs that drive learners (Deci & Ryan, 2015), while the flow experience describes the optimal state that learners can achieve during the learning process (Csikszentmihalyi, 1990; Wang et al., 2021). By leveraging the capabilities of GenAI, GAI-IDLE has the potential to satisfy learners' basic psychological needs as described by SDT, thereby facilitating the emergence of the flow experience (Gao et al., 2025). However, the specific mechanisms through which GAI-IDLE may foster the flow experience based on SDT remain underexplored. Understanding whether these mechanisms exist is essential for optimizing the learning experience, advancing educational technology, and improving learning outcomes. Nevertheless, it is important to remain vigilant, as errors in the data used by GenAI tools could lead to incorrect English learning. Students should approach the information with caution to identify potential mistakes.

Fig. 1 visualizes the potential relationships between GAI-IDLE, basic psychological needs, and flow experience. This study aims to explore these connections as outlined in our proposed model among Chinese college EFL learners. To avoid bias, we also included GenAI literacy, English level, and gender in our model as covariates. Specifically, we formulated the following hypotheses:

H1. GAI-IDLE significantly and positively predicts flow experience.
H2. GAI-IDLE significantly and positively predicts autonomy.
H3. GAI-IDLE significantly and positively predicts competence.
H4. GAI-IDLE significantly and positively predicts relatedness.
H5. Autonomy significantly and positively predicts flow experience.
H6. Competence significantly and positively predicts flow experience.
H7. Relatedness significantly and positively predicts flow experience.

2. Methodology

2.1. Participants

To enhance the diversity of our participant sample, we employed a convenience sampling approach. In collaboration with colleagues involved in English teaching at Chinese universities, we distributed surveys to a broad spectrum of students from various majors who have used GenAI for IDLE. As a result, participants from seven universities across six provinces in China—Zhejiang, Hunan, Henan, Jilin, Guangxi, and Jiangsu—took part in an online survey conducted via the Wenjuanxing platform. The recruitment criterion was that participants must have used at least one GenAI tool for IDLE. A total of 577 questionnaires were collected; however, 244 were excluded due to identical responses or incomplete data. Ultimately, 333 valid questionnaires were retained. It should be noted that all participants completed the Chinese version of the questionnaire and provided informed consent by signing a consent form before participating in the study. Ethical approval for this study was obtained from the Ethics Committee of Hunan Normal University. The demographic characteristics of these participants are provided in detail in Table 1.

2.2. Instruments

2.2.1. GAI-IDLE

GAI-IDLE was assessed using a 8-item unidimensional scale from Liu et al. (2025), with responses measured on a 6-point Likert scale from 1 (strongly disagree) to 6 (strongly agree). An example item is "*I use new AI technologies to find and obtain English learning resources*". A higher score indicates a greater level of diversity and frequency of GAI-IDLE. The detailed items could be seen in Appendix.

2.2.2. Basic psychological needs

Adhering to back-translation procedures (Klotz et al., 2023), we adapted the 9-item three-factor scale from Bergdahl et al. (2023) to assess autonomy, competence, and relatedness, with each factor containing 3 items. Specifically, all items were rephrased to explicitly reference GenAI interactions. For example, an item for autonomy "*I spend more time on the new technologies than I feel I should*" in the original scale was adapted into "*I spend more time on GenAI-assisted English learning than I feel I should*". The same strategy was applied to adapt items for competence and relatedness. Responses were recorded on a 6-point Likert scale, ranging from 1 (strongly disagree) to 6 (strongly agree). The competence items were reversely scored. A higher score indicates greater satisfaction of these basic needs in GAI-IDLE.

2.2.3. Flow experience

Flow experience was assessed using a 10-item unidimensional scale adapted from Wang and Huang (2022), which originally had 14 items designed for assessing flow experience in a blended EFL learning context. Given the differences in contexts, we modified the scale

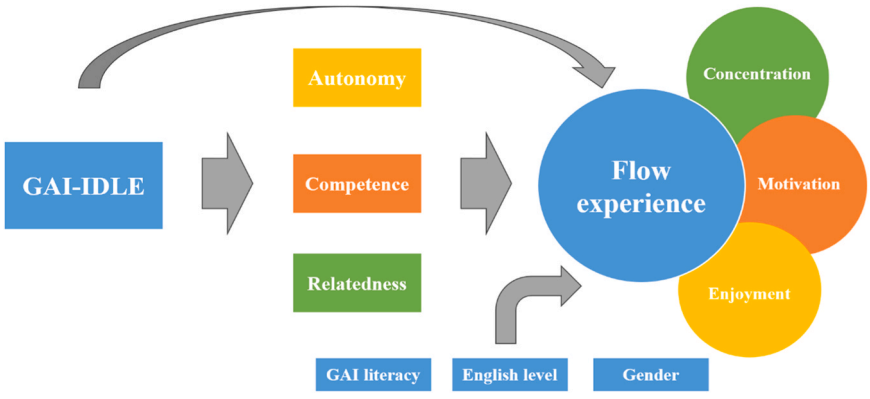


Fig. 1. The SDT-based proposed model of associations between the constructs.

Table 1
Participant demographic details ($n = 333$).

Variable	Category	Count	Percentage
Gender	Male-	103	30.93 %
	Female	230	69.07 %
Age	18–20	211	63.36 %
	21–24	122	36.64 %
Grade	Freshman	139	41.74 %
	Sophomore	114	34.23 %
	Junior	38	11.41 %
	Senior	42	12.62 %

using the Delphi method (Dragostinov et al., 2022). Following the methodologies of McKenzie et al. (1999), which recommend input from at least five experts for content validity, we conducted an in-depth evaluation with a panel of five experts from diverse fields: a computer science professor, a technology education specialist, an assessment expert, and two psycholinguists. We first adapted the items to the GenAI context, and after discussions, the experts advised changing semantically negative items to positive ones to reduce cognitive load for participants. We also found considerable overlap among several items, leading to the removal of four. As a result, 10 items were retained.

Due to the substantial modifications, this process can be regarded as scale development. Following Boateng et al. (2018), we performed an exploratory factor analysis (EFA) using the Maximum Likelihood method with Promax Rotation, retaining factors with eigenvalues greater than 1 (Morrison, 2009). The results indicated a KMO measure of 0.927 and a significant Bartlett's test of sphericity ($\chi^2 = 2621.412$, $df = 45$, $p < 0.001$), confirming the data's suitability for EFA. The analysis revealed that all 10 items loaded onto a single factor, accounting for 67.073 % of the total variance. Factor loadings ranged from 0.725 to 0.850, well above the 0.40 exclusion criterion (Boateng et al., 2018). We did not conduct confirmatory factor analysis (CFA), as its construct validity will be assessed in Section *3.1 Measurement model testing*. Responses were collected using a 6-point Likert scale (1 = strongly disagree, 6 = strongly agree). An example item is "*With the help of GenAI, I can find new ways to make learning English more enjoyable and effective*". Higher scores reflect a more positive flow experience in GAI-IDLE.

2.2.4. GenAI competence

As no existing scale in the literature specifically assesses GenAI literacy, we employed a self-reported method to evaluate participants' GenAI literacy. The statement "*My self-reported GenAI literacy*" was used as the criterion. A 6-point Likert scale was applied, with 1 indicating "very poor" and 6 indicating "very good", where higher scores reflected greater literacy in GenAI.

2.2.5. GenAI literacy

Due to the participants being drawn from various provinces across China, each with different schedules, it was challenging to assess their English proficiency using standardized methods. Consequently, we opted for a self-reported approach to evaluate their English level. The statement "*My self-reported English level*" served as the criterion, and participants rated their proficiency on a 6-point Likert scale, with 1 indicating "very poor" and 6 indicating "very good". Higher scores reflected a higher level of English proficiency.

2.3. Data analysis

Structural Equation Modeling (SEM) can be conducted using two main approaches: covariance-based SEM (CB-SEM) and Partial Least Squares SEM (PLS-SEM) (Hair et al., 2021). For our analysis, we chose PLS-SEM using Smart PLS 4.0 for two key reasons: (1) PLS-SEM is effective with small sample sizes, and given our study has 333 valid responses, it is well-suited to provide robust results (Hair et al., 2022). (2) Our model involves complex relationships, and PLS-SEM excels in handling intricate models with multiple latent variables and indicators. Following Kline's (2016) two-stage SEM procedure, we began by validating the measurement model to assess construct reliability and validity. Confirmatory factor analysis (CFA) was used to examine factor loadings and confirm construct validity. Convergent validity was assessed through Average Variance Extracted (AVE) and Composite Reliability (CR), while discriminant validity was tested using the Fornell-Larcker criterion (Fornell & Larcker, 1981). Finally, we evaluated the structural model by performing bootstrapping with 5000 iterations to assess the significance of path coefficients (Shrout & Bolger, 2002). The significance of direct paths was determined if the p-value was below 0.05, while indirect paths were considered significant if the 95 % confidence interval (CI) did not include 0 (Hayes, 2013).

3. Results

3.1. Measurement model testing

To evaluate the construct validity of the measurement model, confirmatory factor analysis (CFA) was carried out. In line with the advice provided by Hair et al. (2022), which proposes eliminating items having loadings less than 0.7 in PLS - SEM. As presented in Table 2, the factor loadings of all items varied within the range of 0.762–0.993. Thus, each item for every construct demonstrated strong loadings, lending support to the construct validity of the measurement model.

Convergent validity is gauged through AVE and CR. According to Hair et al. (2010), when the AVE values are above 0.5, it implies a robust correlation. Based on Bagozzi and Yi (1988), CR values exceeding 0.7 are indicative of high reliability. Convergent validity is established when both conditions, namely AVE > 0.5 and CR > 0.7, are met. In the context of this particular study (refer to Table 2), the AVE values were found to be within the range of 0.661–0.799. Significantly, all these values were well above the 0.5 benchmark. Additionally, the CR values spanned from 0.749 to 0.955, surpassing the 0.7 threshold. These outcomes strongly confirm the convergent validity of the constructs under investigation. Furthermore, the Cronbach's α values for each construct in this study were higher than 0.7. This additional result further validates the reliability of our measurement model, providing more evidence for the internal consistency of the measurements obtained (Viladrich et al., 2017).

The Fornell-Larcker criterion was used to assess discriminant validity, relying on the AVE for each construct. The criterion suggests that good discriminant validity is present when the square root of a construct's AVE is greater than its correlations with other constructs. As shown in Table 3, the square root of the AVE for each construct (highlighted in bold) was larger than its correlations with other constructs, thereby confirming the discriminant validity of the measurement model (Fornell & Larcker, 1981).

In summary, the validity and reliability of our measurement model were confirmed through CFA and Fornell-Larcker Criterion testing, setting the foundation for the subsequent testing of our structural model.

3.2. Structural model testing

Once the measurement model was validated, we employed the PLS - SEM approach with the bootstrapping technique. This was done to test our proposed hypotheses. The structural model derived from this analysis is depicted in Fig. 2. It should be emphasized that GenAI literacy, English proficiency, and gender were incorporated as covariates within the model to mitigate potential biases. Thus, we initially assessed the influence of these covariates on the structural model. The findings are detailed in Table 4. Evidently, GenAI literacy ($\beta = 0.163$, $p < 0.05$) and English proficiency ($\beta = 0.104$, $p < 0.05$) exerted a significant and positive impact on flow experience. Conversely, gender ($\beta = -0.122$, $p = 0.015 > 0.05$) had no significant effect.

Besides, out of the seven hypotheses we proposed, five were supported by the data. These included: H1, which suggested that GAI-IDLE significantly and positively predicts flow experience ($\beta = 0.338$, $p < 0.05$); H2, which posited that GAI-IDLE significantly and positively predicts autonomy ($\beta = 0.796$, $p < 0.05$); H4, which hypothesized that GAI-IDLE significantly and positively predicts relatedness ($\beta = 0.740$, $p < 0.05$); H5, which proposed that autonomy significantly and positively predicts flow experience ($\beta = 0.288$, $p < 0.05$); and H7, which suggested that relatedness significantly and positively predicts flow experience ($\beta = 0.165$, $p < 0.05$). However, two hypotheses were not supported: H3, which predicted that GAI-IDLE significantly and positively predicts competence ($\beta = 0.064$, $p = 0.573 > 0.05$), and H6, which predicted that competence significantly and positively predicts flow experience ($\beta = 0.077$, $p = 0.065 > 0.05$).

The indirect effects are summarized in Table 5. The findings reveal a significant overall indirect effect of GAI - IDLE on flow experience ($\beta = 0.357$, [0.234, 0.472]). Specifically, GAI - IDLE demonstrated a significant and positive predictive effect on flow

Table 2
The results of confirmatory factor analysis.

Construct	Item	Loadings	AVE	CR	Cronbach's α
GAI-IDLE	IL1	0.843	0.758	0.955	0.954
	IL2	0.880			
	IL3	0.837			
	IL4	0.891			
	IL5	0.869			
	IL6	0.871			
	IL7	0.885			
	IL8	0.885			
Autonomy	AY1	0.754	0.661	0.749	0.742
	AY2	0.870			
	AY3	0.812			
Competence	CE1	0.770	0.770	0.909	0.905
	CE2	0.855			
	CE3	0.993			
Relatedness	RT1	0.862	0.799	0.876	0.874
	RT2	0.917			
	RT3	0.902			
Flow	FW1	0.794	0.671	0.946	0.945
	FW2	0.811			
	FW3	0.833			
	FW4	0.853			
	FW5	0.762			
	FW6	0.810			
	FW7	0.840			
	FW8	0.862			
	FW9	0.811			
	FW10	0.807			

Table 3
The results of Fornell-Larcker criterion.

Construct	GAI-IDLE	Autonomy	Competence	Relatedness	Flow
GAI-IDLE	**0.870**				
Autonomy	0.796	**0.813**			
Competence	0.064	−0.031	**0.877**		
Relatedness	0.740	0.726	−0.035	**0.894**	
Flow	0.764	0.748	0.089	0.687	**0.819**

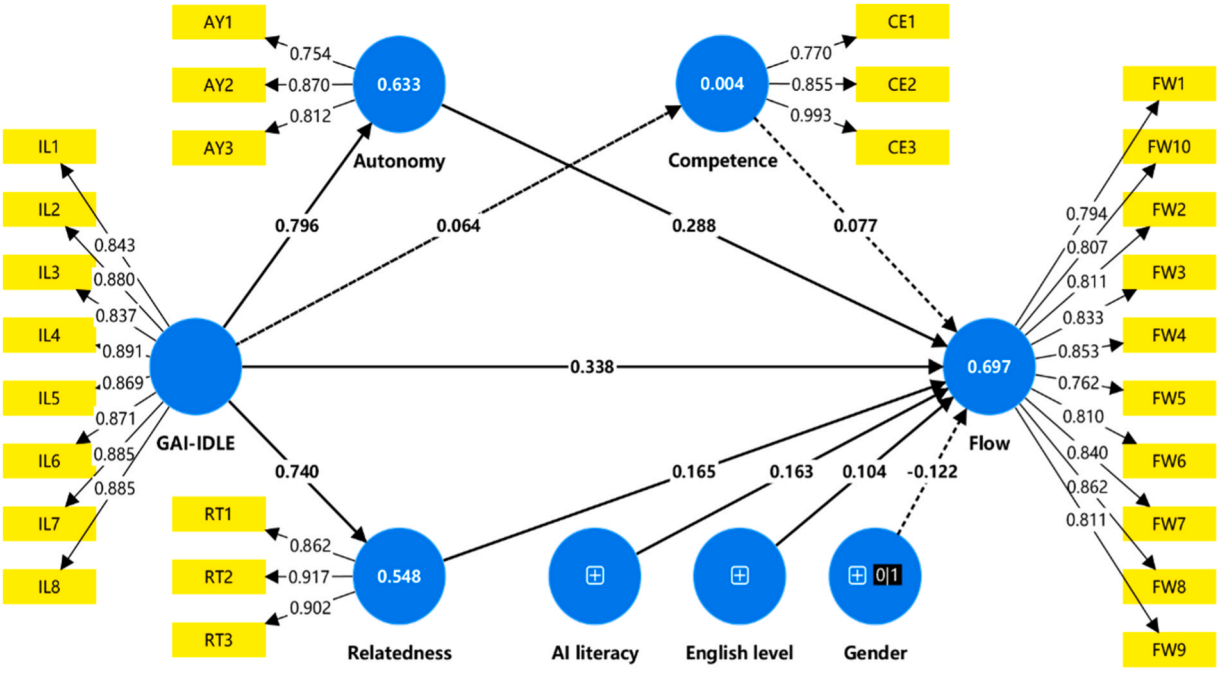


Fig. 2. The structural model.
Note: All coefficients are standardized. The solid lines are statistically significant, whereas the dotted lines are not.

Table 4
The bootstrapping results for direct effects.

Pathway	β	SD	t	*p*	√/×
AI literacy → Flow	0.163	0.038	4.307	0.000	√
English level → Flow	0.104	0.065	4.401	0.015	×
Gender → Flow	−0.122	−0.113	0.067	0.070	×
H1. GAI-IDLE → Flow	0.338	0.074	4.553	0.000	√
H2. GAI-IDLE → Autonomy	0.796	0.029	27.518	0.000	√
H3. GAI-IDLE → Competence	0.064	0.113	0.564	0.573	×
H4. GAI-IDLE → Relatedness	0.740	0.036	20.678	0.000	√
H5. Autonomy → Flow	0.288	0.065	4.401	0.000	√
H6. Competence → Flow	0.077	0.042	1.849	0.065	×
H7. Relatedness → Flow	0.165	0.065	2.541	0.011	√

Table 5
The bootstrapping results for indirect effects.

Pathway	β	SD	95 %CI	Percentage	√/×
Total: GAI-IDLE → Flow	0.357	0.061	[0.234, 0.472]	100 %	√
GAI-IDLE → Autonomy → Flow	0.229	0.054	[0.125, 0.338]	64.15 %	√
GAI-IDLE → Competence → Flow	0.005	0.008	[−0.013, 0.019]	-	×
GAI-IDLE → Relatedness → Flow	0.122	0.047	[0.027, 0.211]	34.17 %	√

experience through autonomy (β = 0.229, [0.125, 0.338]), which accounted for 64.15 % of the total indirect effect. Additionally, GAI - IDLE significantly and positively predicted flow experience through relatedness (β = 0.122, [0.027, 0.211]). However, GAI - IDLE did not show a significant predictive effect on flow experience through competence (β = 0.005, [-0.013, 0.019]).

Notably, autonomy served as a stronger mediator in the relationship between GAI-IDLE and flow compared to relatedness, with its effect size being nearly twice as large. Furthermore, GAI-IDLE, along with the three basic psychological needs, GenAI literacy, English proficiency, and gender, accounted for 69.7 % of the variance in flow experience (see Fig. 2), with English proficiency, gender, and competence being the least significant contributors.

4. Discussion

As the role of flow experience in facilitating effective GAI-IDLE gains increasing recognition, it is imperative to further investigate the underlying mechanisms that shape this experience. Accordingly, this study grounded in SDT, examines how GAI-IDLE, in conjunction with the three basic psychological needs—autonomy, competence, and relatedness—interact to influence flow experience among Chinese college students. Initially, the reliability and validity of the measurement model were established, after which PLS-SEM was employed to analyze the relationships among these variables and test the proposed hypotheses.

To reduce potential bias in our model, we initially assessed the impact of covariates such as GenAI literacy, English proficiency, and gender. The analysis revealed that GenAI literacy was a positive predictor of flow experience, indicating that individuals with greater GenAI literacy were more likely to enter a deeper flow state while engaging with GAI-IDLE. In other words, being familiar with and skilled in GenAI tools seems to promote optimal psychological states during GAI-IDLE activities. To the best of our knowledge, this study is the first to confirm this facilitative effect. This finding aligns with the emphasis on GenAI literacy by Pan and Wang (2025) and Wu and Dong (2025). Additionally, English proficiency was also found to positively predict flow experience, suggesting that individuals with higher English skills may find it easier to immerse themselves in GAI-IDLE activities. This could be due to better comprehension and communication abilities, which increase self-efficacy and facilitate a deeper, more immersive experience (Jia et al., 2024). Surprisingly, gender did not influence the flow experience, contrary to previous studies that highlighted gender differences in technology use (Chen et al., 2024; Liu & Wu, 2024). This could be attributed to the nature of GenAI as an emerging learning tool with interactive and personalized features, which may reduce the impact of gender differences. GenAI's ability to tailor content based on individual learning progress and preferences (Wu et al., 2024) may help create similar flow experiences for students of different genders.

Our proposed hypotheses were largely supported by the findings. Specifically, H1 was confirmed, which aligns with previous experimental studies by Zhang et al. (2024) and Liu (2024). This suggests that GAI-IDLE helps learners maintain focus and immersion by providing personalized content, interactive feedback, and flexible learning methods. Unlike traditional learning environments, it offers a stress-free, personalized experience that reduces anxiety and frustration (Liu et al., 2025). The instant feedback and interactive nature of GAI-IDLE further enhance engagement, motivating learners and promoting a state of "flow" (Wu & Dong, 2025). In addition, H2 and H4 were also supported, indicating that GAI-IDLE nurtures both learners' sense of the ability to make choices and control their learning process and the feeling of connection with others. This might be the first study to explore the effect of GAI-IDLE on fulfilling basic psychological needs. These findings align with the principles of SDT (Deci & Ryan, 2015). In GAI-IDLE, learners might be empowered to choose their pace, topics, and learning methods, which fostered a sense of autonomy and makes them feel more in control of their learning. Moreover, due to teachers' enthusiastic promotion of GenAI and their direct use of it to support teaching, (Derakhshan, 2025), GAI-IDLE likely promoted social interactions through feedback, collaboration, and community features, enhancing learners' sense of relatedness with peers and mentors. This fulfillment of relatedness boosted motivation by meeting learners' needs for connection and support.

Unexpectedly, GAI-IDLE had no significant effect on competence, leading to the rejection of H3. This contradicted SDT, which suggests that a learner's sense of competence is shaped by interactions with the learning environment, particularly when personalized feedback and mastery promotion are involved (Deci & Ryan, 2015). One possible explanation is that our competence-related items primarily focus on concerns about the decline in language learning competence caused by GenAI. Learners may have become overly reliant on GenAI, which, although not harmful to short-term English proficiency as shown in month-long studies (C. Wang et al., 2024; Zhang, Zou, & Cheng, 2025), could lead to worries about the long-term effects of such dependency. This reliance might hinder cognitive processes like memory retention, creativity, and intelligence, as Sternberg (2024) warns, ultimately affecting long-term proficiency development. Similarly, the rejection of H6 was also inconsistent with SDT. According to SDT, competence should naturally foster feelings of mastery and achievement, which drive engagement and immersion (Deci & Ryan, 2015). The rejection of H6 suggests that learners did not feel sufficiently challenged or supported in developing their competence. This could also be due to concerns about the potential negative effects of over-relying on GenAI, which may make them feel that their English proficiency was stagnating, thereby disrupting the flow experience. Consequently, the mediating role of autonomy between GAI-IDLE and flow experience became insignificant.

Furthermore, it was found that autonomy and relatedness significantly and positively predicted the flow experience, thereby supporting H5 and H7. This finding underscores the critical role of autonomy and relatedness in promoting the flow experience within the context of GAI-IDLE. Specifically, when learners experience a sense of control over their learning process and establish a connection with the learning community, they are more likely to achieve deep engagement and immersion. In the context of GAI-IDLE, this suggests that the system's design, which facilitates opportunities for self-directed learning and fosters a sense of connection, can substantially enhance the flow experience. This supports the premise that satisfying learners' psychological needs, as articulated in SDT (Deci & Ryan, 2015), is fundamental to cultivating an optimal state of focus and enjoyment, which, in turn, contributes to

improved learning outcomes. Additionally, these findings align with previous research in other technology-assisted learning contexts (Lüking et al., 2023; Zhai et al., 2024). Given the previously discussed significant effects of GAI-IDLE on autonomy and relatedness, the mediating roles of both factors in promoting flow experience were naturally affirmed.

It is important to note that autonomy emerged as a stronger mediator in the relationship between GAI-IDLE and flow compared to relatedness, with its effect size being nearly twice as large. In the context of GAI-IDLE, this suggests that learners' sense of control and independence in their learning process plays a more significant role in promoting deep engagement and immersion than their sense of connection to the learning community. Additionally, GAI-IDLE, along with the three basic psychological needs, GenAI literacy, English proficiency, and gender, collectively accounted for 69.7 % of the variance in flow experience. This indicates that these factors collectively provide a robust explanation for the variation in flow experience among individuals, leaving only approximately 30 % of the variance unexplained. These findings underscore the strong explanatory power of SDT in elucidating the mechanisms through which GAI-IDLE fosters flow experience.

5. Implications and limitations

This study makes several important theoretical contributions by integrating SDT into the context of GAI-IDLE. It deepens our understanding of how GAI-IDLE, in conjunction with the three basic psychological needs—autonomy, competence, and relatedness—shaped the flow experience for learners. Specifically, the study is one of the first to empirically demonstrate the positive impact of GenAI literacy on the flow experience, extending the SDT framework by introducing GenAI literacy as a key predictor of psychological engagement. Furthermore, the findings contradicted traditional SDT assumptions by showing that GAI-IDLE did not significantly affect competence, nor did competence mediate the relationship between GAI-IDLE and flow experience. This suggests that, in technology-mediated learning environments, competence may play a less central role compared to autonomy and relatedness. The study also emphasizes the critical mediating roles of autonomy and relatedness, with autonomy emerging as a stronger mediator. This underscores the importance of self-directed learning and control in promoting deep engagement and immersion. These insights offer a more nuanced understanding of how psychological needs function in technology-enhanced learning environments.

From a practical standpoint, the findings offer valuable insights for educators, policymakers, and developers of GAI-IDLE platforms. Given the positive impact of GenAI literacy on the flow experience, educational institutions should prioritize incorporating GenAI literacy training into curricula through workshops, tutorials, and hands-on sessions. This would equip students with the necessary skills to engage more effectively with GenAI tools. Moreover, the study highlights the importance of autonomy in enhancing the flow experience, suggesting that GAI-IDLE platforms should emphasize personalized learning environments. These environments should allow learners to control their pace, select topics of interest, and choose learning methods that suit their preferences. Adaptive learning algorithms could be used to tailor content according to individual progress and preferences, further strengthening the sense of autonomy. While autonomy emerged as a more influential factor, relatedness also plays an important role in fostering flow. Thus, GAI-IDLE platforms should integrate features that encourage social interactions, such as collaborative projects, discussion forums, and peer feedback mechanisms, to help learners feel part of a community. However, the study also raises concerns about the potential drawbacks of over-reliance on GenAI, especially in terms of long-term proficiency development. Educators should design GAI-IDLE activities that strike a balance between using GenAI and traditional learning methods to ensure the cultivation of essential cognitive skills such as memory retention, creativity, and problem-solving.

Despite its valuable contributions, the study has several limitations that should be addressed in future research. The sample was restricted to Chinese college students, which may limit the generalizability of the findings. Future studies should incorporate more diverse samples, including participants from different age groups, cultural backgrounds, and educational levels, to confirm the results in a broader range of contexts. Additionally, the cross-sectional design used in this study limits the ability to assess long-term effects. Longitudinal research is necessary to determine whether the initial benefits of GAI-IDLE, such as enhanced flow experiences, are sustained over time and how they affect long-term proficiency development. Lastly, the study raises concerns about the potential negative consequences of over-relying on GenAI, such as impaired cognitive processes and stagnation in skill development. Future research should explore strategies to mitigate these effects, such as integrating metacognitive training or designing activities that foster critical thinking and independent problem-solving. Comparative studies evaluating the effectiveness of GAI-IDLE versus other forms of technology-mediated learning or traditional classroom instruction could also provide valuable insights into its unique strengths and limitations. By addressing these limitations and exploring these future research directions, scholars can refine the theoretical framework of SDT within the context of GAI-IDLE and develop more effective strategies to enhance the flow experience and improve overall learning outcomes.

6. Conclusion

Building upon SDT, this study explored the mechanisms underlying the relationships between GAI-IDLE, the three basic psychological needs, and flow experience among Chinese college students. The results indicated that, regarding covariates, GenAI literacy and English level had a direct, positive impact on flow experience, whereas gender did not exert a direct effect. Concerning the core variables, GAI-IDLE was found to directly and positively predict flow experience, as well as autonomy and relatedness, but did not directly influence competence. Furthermore, autonomy and relatedness were found to directly and positively affect flow experience, whereas competence did not have a direct effect. Additionally, GAI-IDLE indirectly and positively predicted flow experience through the mediation of autonomy or relatedness, but not through competence.

Consent to participate

Informed consent to participate was obtained from all individual participants included in the study.

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CRediT authorship contribution statement

Wang Yongliang: Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Wu Hanwei:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no competing interests.

Appendix

GAI-IDLE (8 items)

1. I engage in English conversations with AI-powered chatbots on various topics to increase my exposure to English.
2. I use AI-powered speech recognition tools to practice speaking English and receive immediate feedback on my fluency and intonation.
3. I play AI-powered language learning games to expand my English vocabulary.
4. I interact with AI-powered chatbots to acquire personalized English learning tips and strategies.
5. I use new AI technologies to find and obtain English learning resources.
6. I engage in discussions with AI-powered chatbots to deepen my understanding and knowledge of English language culture.
7. I interact with AI-powered chatbots to practice my English writing skills beyond the classroom.
8. I use AI technologies to simulate real-life English language use situations.

Basic psychological needs (9 items)
Autonomy:

1. I spend more time on GenAI-assisted English learning than I feel I should
2. GenAI-assisted English learning ends up making me complete some learning tasks I did not want to do.
3. GenAI-assisted English learning intrude into my daily life.

Competence:

4. Using GenAI to learn English has made me feel somewhat insecure about my language abilities.
5. Using GenAI to learn English has made me feel that my language learning skills have diminished.
6. Using GenAI to learn English has made me lose some of my confidence.

Relatedness:

1. Using GenAI to learn English has helped me feel a greater sense of belonging to the community of English learners.
2. Using GenAI to learn English has helped me feel close and connected with other people who are important to me.
3. Using generative AI to learn English, I feel closer to some others.

Flow experience (10 items)*:*

1. With the help of GenAI, I can learn English anytime and anywhere, improving my language skills and maintaining my interest.
2. I enjoy making full use of GenAI's features in my English learning, enabling me to complete tasks more efficiently and deepen my understanding of the language.
3. With GenAI's support, I can confidently face challenging English tasks and gradually improve my abilities.
4. Whenever GenAI participates in my English learning, its intelligent feedback and real-time guidance make me feel that everything is progressing in an organized way.
5. Even when English learning tasks are not mandatory, I am eager to learn on my own, especially with GenAI's instant support.
6. With the help of GenAI, I can find new ways to make English learning more interesting and productive.

7. With GenAI's support, I am able to focus more and complete tasks more efficiently, while deepening my understanding of English.
8. English learning tasks are both challenging and stimulating for me, especially with GenAI's help, making it easier to tackle challenges.
9. With GenAI's support, I can handle high-demand tasks with greater composure and alleviate stress through proper planning.
10. Every time I am guided by GenAI, I can clearly understand the next steps in my learning and smoothly complete tasks.

Data availability

Data will be made available on request.

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